

Distribution Statement IRA-6876

EQUIPMENT SPECIFICATION

FOR

VKF PLANT INTERSTAGE COOLER BUNDLES ASSEMBLIES

ARNOLD ENGINEERING DEVELOPMENT COMPLEX

ARNOLD AIR FORCE BASE, TENNESSEE 37389-9998

TABLE OF CONTENTS

1 SCOPE

1.1 Summary

1.2 Background

2 APPLICABLE DOCUMENTS

2.1 Government Documents

2.2 Non-Government Documents

3 REQUIREMENTS

3.1 General

3.2 Workmanship

3.3 Performance

3.4 Physical Requirements

3.5 Mechanical

3.6 Marking

3.7 Cleaning

3.8 Verification and Acceptance

3.9 Packaging

3.10 Submittals

Appendix A – Reference Drawings

Appendix B – Performance Data

Appendix C – Submittal Procedure

1. SCOPE

- 1.1 Summary: The Contractor shall supply 5 cooler bundle assemblies. Each assembly shall consist of all tubes, any supports, and fins as needed. These cooler bundles will be installed by the Government at the Arnold Engineering Development Complex (AEDC) Arnold Air Force Base.
- 1.2 Background: The first stage coolers (X11, X12, X13, X14) and the Second Stage coolers (X21 and X22) use water in the tubes to cool the air and are of fin tube construction. X71, X81, and X91 use water in the shell to cool air in the tubes.

2. APPLICABLE DOCUMENTS

2.1 Government Documents:

2.1.1 Reference Drawings (Appendix A):

- A. 441-M2 – Gas Dynamics Facility Horizontal Intercoolers Details & Controls Dimensions Specification 2-B-14.
- B. MV-7153 – X11, X12, X13, X14 Intercooler.
- C. VU11338 – Tube Bundle Details- Basco First Stage Cooler X11, X12, X13, X14.
- D. VU114478 Tube Bundle Details- AO Smith First Stage Cooler X11, X12, X13, X14.
- E. MV-7154 – X21, X22 Intercooler.
- F. VU11338 – Tube Bundle Details – Basco Second Stage Cooler X21 & X22.
- G. MV-7166 – X71 Intercooler.
- H. MV-7165 – X81 Intercooler.
- I. HE-70194 – X91 Aftercooler (Shell only no bundle).

2.2 Non-Government Documents:

2.1.1. American Society of Mechanical Engineers (ASME):

- | | | |
|----|---------------|----------------------------------|
| A. | ASME Y14.5-18 | Dimensioning and Tolerancing. |
| B. | ASME BPVC-23 | Boiler and Pressure Vessel Code. |

2.1.2. American Institute of Steel Construction (AISC).

A. AISC-ASD Allowable Stress Design.

3. REQUIREMENTS

3.1 General:

3.1.1 For the first stage bundles (X11, X12, X13, X14) and second stage bundles (X21 and X22) shall be of fin tube construction. Cooler construction shall be a 4-pass triple inlet and outlet waterflow style cooler. Each bundle shall be designed, manufactured, and verified in accordance with all requirements in ASME BPVC.

3.1.2 X71, X81, & X91 bundles shall be of shell and tube construction. Cooler construction shall be a 2-pass single inlet & outlet waterflow cross baffled style bundle. Each bundle shall be designed, manufactured, and verified in accordance with all requirements in ASME BPVC.

3.2 Workmanship:

3.2.1 All components of each bundle shall be constructed and finished in a manner such that no sharp edges, burs, and physical defects are present.

3.3 Performance:

3.3.1 Each fin and tube bundle shall be designed to operate at required capacity with up to 20% of the tubes plugged.

3.3.2 The new bundles shall meet or exceed all performance requirements outlined in Appendix B.

3.4 Physical Requirements

3.4.1 Each bundle shall be designed to fit within the existing envelope without modification or addition to the existing structures that house each bundle. The bundle's overall width, height, and length shall not exceed what is shown in the reference drawings. Drain connections, supports, and any interfaces between the bundle and shell or support structure shall maintain existing dimensions.

3.3.2 The bundles shall be designed such that they are supported by the tube sheet using existing mounting points within the shell as shown on the reference drawings. The new bundles shall not exceed the weight of existing bundles, as shown on the drawings.

3.4.2 Each of the X11, X12, X13, X14, X21, & X22 bundles shall have rigging points that support the entire weight of the cooler for lifting with a crane. Rigging points shall be designed using AISC-ASD. At least two rigging points on the bottom and two rigging points on top shall allow for the lifting and rotation of these bundles.

The rigging points on top shall not extend past the top edge of the cooler. The bottom mounted anchor points may increase the profile of the cooler up to 5". Submit a lift diagram with capacity of each rigging point.

- 3.4.3 Each of the X71, X81, & X91 bundles shall have removable studs to allow for screw in anchor points at the tube sheet. Rigging points shall be designed using AISC-ASD. These shall allow for removal of the bundle from the cooler. An additional rigging point to rotate the bundle shall be provided at the bottom quarter of the cooler that will not expand the profile of the bundle. Supports to reinforce these anchor points may be added as necessary as long as they do not interfere with the existing configuration. Submit a lift diagram with capacity of each rigging point.

3.5 Mechanical:

- 3.5.1 Each of the bundles shall interface with all existing drains and flex hoses for each bundle found on reference drawings in Appendix A. Design conditions are provided in Appendix B.
- 3.5.2 Do not use lead or asbestos materials in new components.
- 3.5.3 The bundles shall be constructed of a material to withstand the following cooling water properties:
- A. pH: 6.7 to 7.7.
 - B. Suspended Solids: 4 to 20 parts per million (ppm).
 - C. Dissolved Solids: 70 to 130 ppm.
 - D. Silica: 1 to 10 ppm.
 - E. Calcium: 19 to 29 ppm.
 - F. Bromine: 6.5 ppm.
 - G. Bicarbonate: 64 to 70 ppm.
 - H. Hardness as CaCO_3 : 67.
 - I. Specific Conductance: 140 to 172.

3.6 Marking

- 3.6.1 Each assembly will be marked with the intercooler bundle name and the upstream and downstream side.
- 3.6.2 Permanently attached nameplates shall be provided showing at a minimum:

- A. Manufacturer name.
- B. Maximum allowable working pressure.
- C. Serial number.
- D. Manufacturer drawing number.
- E. Test pressure.
- F. Number of tubes.
- G. Weight.

3.6.3 Nameplate shall not interfere with operation and be visible when the bundle is removed from the shell.

3.7 Cleaning:

3.7.1 After tube installation into the bundle assembly, each bundle shall be free of oxides, oils, and any lubricants used in aiding penetration of the tube sheet. In addition, tubes shall be clean. All tube sheet holes shall be cleaned by a carbide ball bur that is 0.010"-0.015" larger than the nominal tube diameter.

3.8 Verification, Testing and Acceptance

3.8.1 Engineering Analysis: The contractor shall calculate performance for each fin and tube cooler design. The performance estimates shall include efficiency estimates, water and air temperature estimates, and performance at full capacity and with 20% of the tubes plugged. Provide the following:

- A. Stress analysis including all supporting information, required to verify all major components of the cooler under load, including, but not limited to, the bundle tubes, fasteners, welds, and baffles are suitable for all design conditions identified in Appendix B Performance Data. Evaluated loads shall include dynamic loads created by air flow at the identified operating conditions.
- B. Analysis, including all supporting information, required to verify that the flow characteristics meet the requirements of Appendix B Performance Data.
- C. Analysis, including all supporting information, required to verify that the performance characteristics meet the requirements of Appendix B Performance Data.
- D. Final analysis shall be submitted in a single complete and certified package. Final approval will only be for items provided within this

package. The package shall also certify that all analysis provided is accurate and consistent with the associated final drawings, and which drawings are applicable to each analysis item.

3.8.2 Testing: All testing shall be performed at the Contractor's facility. Government reserves the right to witness all tests.

- A. Pressure tests: Each of the X11, X12, X13, X14, X21, & X22 interstage bundles shall be factory tested with a full vacuum. Any leaks found shall be repaired at the by the contractor, at no additional cost to the Government, and subsequently retested.
- B. X71 tubes shall be hydro tested per ASME BPVC at the factory before arrival to AEDC. Any bundles discovered leaking will be repaired by the contractor, at no additional cost to the Government, and subsequently retested.
- C. X81 tubes shall be hydro tested to per ASME BPVC at the factory before arrival to AEDC. Any bundles discovered leaking will be repaired by the contractor, at no additional cost to the Government, and subsequently retested.
- D. X91 tubes shall be hydro tested per ASME PBVC at the factory before arrival to AEDC. Any bundles discovered leaking will be repaired by the contractor, at no additional cost to the Government, and subsequently retested.
- E. Fabrication inspection: All critical dimensions such as overall dimensions, and dimensions of interface points shall be documented and submitted. Welding inspections shall be submitted per ASME BPVC.

3.9 Packaging

3.9.1 The Contractor shall provide all preservation and packing to ensure safe delivery of the interstage bundles to AEDC. All vents/drains and tube openings shall covered/plugged to prevent contamination. Bundles shall be packaged for long term outside storage at AEDC.

3.9.2 Delivery Condition: bundles shall be supplied in a dry and undamaged condition.

3.10 Acceptance: Acceptance shall occur upon receipt and approval of all submittals, and receipt of the cooler arriving without damage to Arnold Engineering Development Complex (AEDC) Arnold Air Force Base.

3.11 Submittals – Provide the following in accordance with Appendix C, Submittal Procedure. Provide the following submittals:

- 3.11.1 Analysis and design: The Contractor shall provide analysis and calculations to verify the cooler meets the requirements of this specification.
- 3.11.2 Fabrication Drawings: The Contractor shall submit one 11-inch by 17-inch hard copy and one electronic set of detailed fabrication drawings or Government approval prior to fabrication in accordance with ASME Y14.5 and Appendix C – Submittal Procedure. Include detailed material specifications for all materials and components.
- 3.11.3 Testing Reports: Reports for all hydro tests and vacuum checks shall be submitted prior to shipment.
- 3.11.4 Inspection reports for any machining and fabrication shall be submitted prior to delivery.
- 3.11.5 Warranty: Contractor shall warranty work against defects in workmanship and materials for one year following acceptance or provide manufacturer's standard warranty, whichever is longer.
- 3.11.6 Required submittal summary table:

Table I.

No.	Section Reference	Submittal Description	Due
1	3.4.2	Lift diagram for X11, X12, X13, X14, X21, & X22 bundles	Before fabrication
2	3.4.3	Lift diagram for X71, X81, & X91 bundles	Before fabrication
3	3.8.1, 3.11.1	Engineering Analysis	Before fabrication
4	3.11.2	Fabrication Drawings	Before fabrication
5	3.8.2, 3.11.3	Testing Reports	Prior to shipment
6	3.8.2.E, 3.11.4	Inspection Reports	Prior to delivery

APPENDIX A
REFERENCE DRAWINGS

DRAFT

APPENDIX B
PERFORMANCE DATA

Table 2 Air Flow conditions for pressure drop across coils:

Bundle	Mass Flow of air (#/sec)	Inlet air Temp (°F)	Inlet operating pressure (psia)	Inlet Design pressure (psia)	Max Rate of Inlet Air Temp Change (°F /min Increase or Decrease)	max Outlet air Temp, (°F)
X11,12,13,14	526	222	45	52	200	100
X21,22	438	222	45	65	200	100
X71	237	400	1600	1600	200	150
X81	200	400	2500	2500	200	150
X91	125	308	2774	4200	200	150

Δp shall not exceed 2% of p_{inlet}

Table 3 Water Flow conditions:

Bundle type	Max water Pressure (psig)	Normal Water Pressure (psig +/- 20%)	Max. Inlet Water Temp, (°F)	Max. outlet Water Temp, (°F)	Max. water flow (GPM) at maximum temp
X11,12,13,14	150	100	80	140	3520
X21,22	150	100	80	140	3660
X71	140	100	80	140	1560
X81	140	100	80	140	1355
X91	140	100	80	140	1100

The tubes shall have an assumed fouling factor of $.0020 \frac{hr*ft^2*F}{BTU}$

END OF APPENDIX B

APPENDIX C
SECTION 01 33 00
SUBMITTAL PROCEDURE

PART 1 GENERAL

- 1.1 DESCRIPTION OF REQUIREMENTS. This section specifies procedural requirements for non-administrative submittals, including but not limited to shop drawings, product data, manufacturer's certificate, design data, calculations, and verifications, manufacturer's instructions, manufacturer's field service reports, samples, operation and maintenance manuals, and other miscellaneous work-related submittals. These submittals are required to amplify, expand, and coordinate other information contained in the contract. This section describes the type and content of submittals that may be required for this work. The required submittals are included in the Statement of Work.

Non-work-related submittals are addressed elsewhere in the contract rather than in the specification and may include items such as: contract progress schedule, permits, payment applications, performance and payment bonds, insurance certificates, and progress reports.

The Contracting Officer may request submittals in addition to those specified when deemed necessary to adequately describe the work covered in the respective sections of the Statement of Work.

1.2 SUBMITTAL PROCEDURE

- A. Definition: For the purposes of this section, the term "Contractor" denotes the prime Contractor.
- B. Listing: At the end of this section is a summarized listing of item submittals requiring approval for the work. The listing is included for the convenience of users of the contract documents. The listing may not be all inclusive and additional item submittals may be required. Within 14 calendar days after receiving contract award, Contractor shall submit a copy of the summarized submittal listing with calendar dates assigned to each item submittal indicating when submittal will be received by the Government. If contract requires a pre-construction conference, Contractor may submit completed submittal listing at the conference.
- C. Risk: Do not proceed with the part of the work covered by an item submittal including purchasing, fabricating, and delivering until approval is received. Machinery, equipment, material, and articles that do not have the required approval shall be installed or used at the risk of subsequent rejection. Any fabrication or other work performed in advance of the receipt of accepted item submittals and approvals shall be entirely at the Contractor's risk and expense.

- D. Transmittal Timing: Coordinate the preparation and processing of item submittals with the performance of the work. Prepare and transmit each item submittal to the Contracting Officer sufficiently in advance of the performance of related work and other applicable activities. Transmit related item submittals for the same unit of work so that processing will not be delayed by the Government's need to review submittals concurrently for coordination. No delay damages or time extensions will be allowed for time lost in late submittals.
- E. Content: Each submittal shall be complete and in sufficient detail to allow ready determination of compliance with contract requirements. All submittals which are generic and list more information than is specifically required shall be marked to identify required information. Complete a material approval submittal and attach as cover sheet for each submittal. Contractor may include multiple item numbers on a material approval submittal.
- F. Language: All item submittals shall be written in English, including documents, notes on drawings, sketches, and/or samples, calculations, manuals, and all other instances of written text and communication.
- G. Units: All item submittals shall be marked or show dimensions and values in the same units as specified in the contract documents.
- H. Review Time: Allow sufficient time so that contract performance will not be delayed as a result of the time required to properly process submittals, including time for re-submittals, if necessary. Allow 14 calendar days for initial Government processing of each submittal. No extension of time will be authorized because of the Contractor's failure to transmit submittals to the Government sufficiently in advance of the work.
- I. Contractor Certification: All submittals shall be carefully reviewed by an authorized representative of the Contractor prior to submission to the Government. Each submittal shall be dated, signed, and certified by the Contractor as being correct and in strict conformance with the contract documents. No consideration for review by the Government of any Contractor's submittal will be made for any items which have not been so certified by the Contractor. All noncertified submittals will be returned to the Contractor without action taken by the Government, and any delays caused thereby shall be the total responsibility of the Contractor.
- J. Deviations: Should any item submittals required by the contract documents show deviations from the contract requirement, the Contractor shall make specific mention of such deviations in the letter of transmittal, including stating cost effects, and product and system limitations which may adversely affect the work, in order that if acceptable, suitable action may be taken for proper adjustment of the contract; otherwise the Contractor will

not be relieved of the responsibility for executing the work in accordance with the contract documents and the approved submittals. Contractor shall clearly mark the proposed variation in all documentation and specifically point out deviations from contract requirements in transmittal letters. Failure to point out deviations may result in the Government requiring rejection and removal of such work at no additional cost to the Government. Deviations from contract requirements require Government approval and will only be considered when advantageous to the Government. When proposing deviations, deliver written request to the Contracting Officer, with documentation of the nature and features of the deviation and why the deviation is desirable and beneficial to the Government. If lower cost is a benefit, include an estimate of the cost savings. In addition to documentation required for deviation, include the submittal information required for the item. Allow an additional 14 calendar days beyond normal submittal review period for consideration by the Government of submittals with deviations.

- K. Approved Submittals: The part of the work covered by the approved item submittal may proceed provided it complies with the requirements of the contract documents. Final acceptance will depend upon that compliance. The term "Approved" shall only indicate that there is no exception taken to the submittal. Approval of the item submittal shall not be construed as a complete check, and indicates only that the general method or other information appears to meet the contract requirements. Approval does not relieve the Contractor of the responsibility for any error which may exist. After item submittals have been approved by the Contracting Officer, no resubmittal for the purpose of substituting materials or equipment, or changes to any other information will be considered unless accompanied by an explanation of why a substitution or change is necessary.
- L. Disapproved Submittals: Contractor shall correct disapproved item submittals and resubmit for approval within timeframe noted. If no date is given, Contractor shall submit corrected submittal within 14 calendar days. If the Contractor considers any correction or notation on the returned submittal to constitute a change to the Contract Drawings, Specification or Statement of Work the Contractor shall notify the Contracting Officer.
- M. Responsibility: The Government's review of Contractor submittals shall not relieve the Contractor of the entire responsibility for the correctness of details and dimensions and conformance to the specifications. The Contractor shall assume all responsibility and risk for any mistakes and/or costs due to any errors in submittals.
- N. Inconsistencies: If a conflict or inconsistency arises between an approved item submittal and the contract documents, the contract documents shall govern.

1.3 SPECIFIC SUBMITTAL DESCRIPTIONS AND REQUIREMENTS: Submittal requirements for individual units of work are specified in the applicable specification section or Statement of Work (SOW). Except as otherwise indicated in the individual specification sections or SOW, comply with the following requirements for each type of submittal.

- A. Shop Drawings: These are technical drawings and data specially prepared for this contract including fabrication and installation drawings, setting and seaming diagrams, and coordination drawings (for use on-site). Shop drawings include drawings, diagrams and schedules specifically prepared to illustrate some portion of the work, diagrams and instructions from a manufacturer or fabricator for use in producing the product and as aids to the Contractor for integrating the product or system into the project, or drawings prepared by or for the Contractor to show how multiple systems and interdisciplinary work will be coordinated. Information required on shop drawings includes dimensions, identification of specific products and materials which are included in the work, compliance with specified standards, and notations of coordination requirements with other work. Provide special notations of dimensions that have been established by field measurements. Highlight, encircle, or otherwise indicate deviations from the contract documents on the shop drawings. Furnish one hard copy and one electronic copy in AutoCAD version 2020 or earlier drawing file format (.dwg). The only permissible text font is ROMANS. Custom fonts and shapes are prohibited. The minimum allowable font size shall be 0.125-inch. The maximum allowable font size shall be 0.25-inch.
- B. Product Data: This data includes standard printed information on manufactured products that has not been specially prepared for this contract, including manufacturers' product specifications illustrating size, physical appearance and other characteristics of materials, installation instructions, standard color charts, catalog cuts, illustrations, schedules, standard wiring diagrams, standard product operating and maintenance manuals, and samples of warranty language when the contract requires extended product warranties. General information required specifically as product data includes manufacturers' standard printed recommendations for application and use, compliance with recognized standards of trade associations and testing agencies, the application of their labels and seals (if any), special notation of dimensions which have been verified by way of field measurement, and special coordination requirements for interfacing the material, product, or system with other work. Furnish three hard copies and one electronic copy.
- C. Samples: These are physical examples of work, including, swatches showing color, texture, and pattern, color-range sets, and units of work to be used for independent inspection and testing. Samples include fabricated or unfabricated physical examples of materials, equipment or workmanship that illustrate functional and aesthetic characteristics of a material or product

and establish standards by which the work can be judged, color samples from the manufacturer's standard line (or custom color samples if specified) to be used in selecting or approving colors for the project, and field samples and mock-ups constructed on the project site to establish standards by which the ensuring work can be judged. This includes assemblies or portions of assemblies which are to be incorporated into the project and those which will be removed at conclusion of the work. Submit samples for the Contracting Officer's visual review of general kind, color, pattern, and texture for a final check of the coordination of these characteristics with other related elements of the work and for quality control comparison of these characteristics between the final sample submittal and the actual work as it is delivered and installed. Submit one each.

- D. Design Data: Design data includes design calculations, mix designs, analyses or other data pertaining to a part of work. Refer to individual sections of the Specification or Statement of Work for required quantities, formats, and signatures or certifications required. Furnish three hard copies and one electronic copy.
- E. Test Plans and Reports: Test plans shall include planned testing, including a description of the test, equipment and supplies needed, and step-by-step notation of test activities and tasks. Test reports include reports signed by an authorized official of a testing laboratory that a material, product or system identical to the material, product or system to be provided has been tested in accordance with the specified requirements. Testing shall have been within three years of date of contract award for the project. Reports also include findings of tests required to be performed by the Contractor on an actual portion of the work or prototype prepared for the project before shipment to the job site, findings of tests made at the job site or on a sample taken from the job site, investigation reports, daily logs and checklists, and final acceptance test and operational test procedures. Furnish three hard copies and one electronic copy of each such report required.
- F. Certificates: These include statements printed on the manufacturer's letterhead and signed by the responsible officials of the manufacturer of a product, system or material attesting that the product, system or material meets specification requirements. Certificates include documentation required of the Contractor, or of a manufacturer, supplier, installer or subcontractor through the Contractor, to further demonstrate the quality of orderly progressions of a portion of the work by documenting procedures, acceptability of methods or personnel qualifications. Examples include confined space entry permits, and text of posted operating instructions. Certificates shall be dated after award of the project contract and clearly name the project. Furnish three hard copies and one electronic copy of each certificate required.

- G. Manufacturer's Instructions: Preprinted material describing installation of a product, system or material, including special notices and Material Safety Data sheets concerning impedances, hazards and safety precautions. Furnish three hard copies and one electronic copy.
- H. Manufacturer's Field Reports: Documentation of the testing and verification actions taken by a manufacturer's representative at the job site, in the vicinity of the job site, or on a sample taken from the job site, on a portion of the work, during or after installation, to confirm compliance with the manufacturer's standards or instructions. The documentation shall be signed by an authorized official of a testing laboratory or agency and shall state the test results, and indicate whether the material, product, or system has passed or failed the test. Furnish three hard copies and one electronic copy.
- I. Operation and Maintenance Data: Data furnished by the manufacturer, or the system provider, to the equipment operating and maintenance personnel, including manufacturer's help and product line documentation necessary to maintain and install the equipment, and needed by operating and maintenance personnel for the safe and efficient operation, maintenance, and repair of the item. The data is intended to be incorporated in an operations and maintenance manual or control system. Furnish two hard copies and one electronic copy.
- J. Final Design Drawings and Documentation: Final design documentation including drawings, specifications, estimates, and other required deliverables. Electronic documentation submitted shall include every associated component file of drawings and other documentation, including raster images, jpeg, tiff, Excel, and any related items required to view, modify, or manipulate the electronic files. Furnish a CD-ROM of all final design documentation, and drawings in AutoCAD 2020 version or earlier drawing file format (.dwg), to the Contracting Officer for approval prior to applying for final payment. The drawings shall be fully editable using AutoCAD software. In addition, drawings shall be delivered to clearly show the location and arrangement of all materials and equipment. The only permissible text font is ROMANS. Custom fonts and shape files are prohibited. The minimum allowable font size shall be 0.125-inch. The maximum allowable font size shall be 0.25-inch.
- K. Record (As-Built) Drawings: Record drawings showing final configuration of work accomplished. Show all changes, additions, and deviations from the original contract drawings and documentation. If no changes occur, furnish certification to that effect. Drawings shall be redlined electronic drawings, unless waived by the Government, and shall accurately show as-built conditions during the progress of the job. Furnish drawings on a CD-ROM, in AutoCAD 2020 or earlier version, including every associated component file of the drawings, including raster images, jpeg, tiff, Excel, and any related

items required to view, modify, or manipulate the electronic files. Submit to the Contracting Officer for approval prior to applying for final payment.

- L. Miscellaneous Submittals: These are work-related, non-administrative submittals that do not fit in the previous categories, including the following:
1. Maintenance agreements. Furnish one hard copy and one electronic copy.
 2. Survey data and reports. Furnish one hard copy and one electronic copy.
 3. Project photographs. Furnish both hard copies and digital files.
 4. Keys and other security protection devices.
 5. Maintenance tools, spare parts, and overrun or maintenance stock. Refer to individual sections of the specification for required quantities of spare parts, extra and overrun stock, maintenance tools and devices, keys, and similar physical units to be submitted.
 6. Qualification certificates. Furnish one hard copy and one electronic copy.
 7. Employee training certificates and documentation showing successful completion of training.
 8. Documentation of percentage of recovered material content used during contract.
 9. Contractor safety and work plans and schedules.
 10. Warranties.

END OF STATEMENT OF WORK